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SEQUENCE ALIGNMENT

→ The process of arranging 2 or more DNA/RNA/protein sequences to identify regions of similarity

→ It helps us to know:

↳ Are the two genes related?

↳ Did they evolve from a common ancestor?

↳ Do they perform similar functions?

↳ Where are mutations?

Seq 1 → A T G C T A

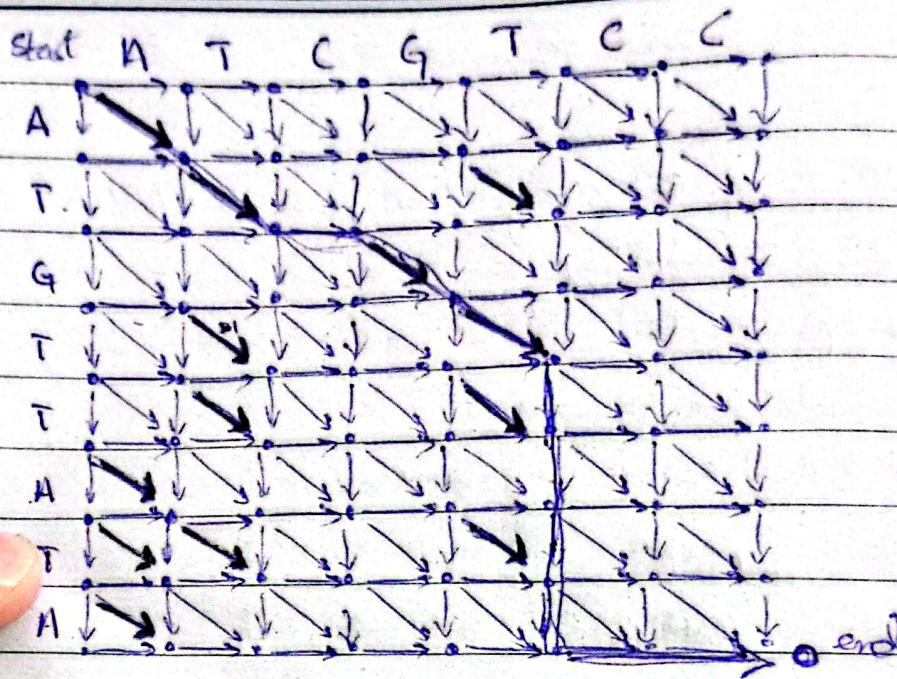
Seq 2 → A T G G T A

we see one diff which can be due to mutation.

↳ This eventually becomes the Longest Common Subsequence Problem:

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Black arrows corresponds to matching symbols.
We have to find the longest path which covers maximum black arrows

- For matching symbols we give range to +1 else 0.
- So total score we get is 4.
- We will use the Dynamic Programming to solve this.

$$S_{i,j} = \max \begin{cases} S_{i+1,j} + 0 \\ S_{i,j-1} + 0 \\ S_{i-1,j-1} + 1 \text{ if } S[i] = S[j] \end{cases}$$

- DP R thru ~~sub~~ Score of the subsequence nikalenge.
- Actual Path hum backtracking k thru nikalenge

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▷ ALIGNMENTS

Global Alignment

- Tries to align entire sequence
- Align all letters from query and target
- Suitable for closely related sequences
- ~~Align~~

Local Alignment

- Align regions having highest similarities
- Align substrings of target with substring of query
- Suitable for more divergent sequences

• Needleman - Wunsch Algo

↳ Algo for Global Alignment

① Initialization

i. $F(0,0) = 0$

ii. $F(0,j) = -j \times d$

↳ All values of 1st row
= index val $\times d$

iii. $F(i,0) = -i \times d$

↳ All values of 1st col
= index $\times d$

Seq. = 1 (j)

	0	1	2	3	4	5
0	-	A	T	G	C	T
1	A	-2				
2	G	-4				
3	C	-6				
4	T	-8				

j+1

$d = 2$ (penalty)

↳ gap penalty

indel penalty

insertion / deletion

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(2) Main Iteration

for each i to $(len(seq)-2)$ for each j to $(len(seq)-1)$

$$F(i,j) = \max \begin{cases} F(i-1,j) - d & \text{case 1} \\ F(i,j-1) - d & \text{case 2} \\ F(i-1,j-1) + S(x_i, y_j) & \text{case 3} \end{cases}$$

matrix score val

$$Pts(i,j) = \begin{cases} \text{UP} & \text{if case 1} \\ \text{LEFT} & \text{if case 2} \\ \text{DIAG} & \text{if case 3} \end{cases}$$

→ jis case ki max value choose krenge wohi arrow yaha choose hoga.

→ Scoring Matrix

	A	T	G	C	T
A	1	-1	-1	-1	-1
G	-1	-1	1	-1	-1
C	-1	-1	-1	1	-1
T	-1	1	-1	-1	1

match = -1

mismatch = 1

→ F

	A	T	G	C	T
A	0	-2	-4	-6	-8
T	-2				
G					
C					
T					

Page No.

- A C G C T G
C A T G - T -

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		A	T	G	C	T
	O	-2	-4	-6	-8	-10
A	-2	1 ^R	-1 [←]	-3 [←]	-5 [←]	-7 [←]
G	-4	-1 [↑]	0 ^R	0 ^R	-2 [←]	-4 [←]
C	-6	-3 [↑]	-2 [↑]	-1 [←]	1 ^R	-1 [←]
T	-8	-5 [↑]	-2 ^R	-3 [↑]	-1 [↑]	-2 ^R

A T G C T

Seq → 1 A T G C T
A - G C T

• Smith-Water Algorithm

- ↳ Used for Local Sequence Alignment
- Almost same as Previous one
- All negative val converted to ZERO

① Initialization

- Fill ~~all~~ columns with 0
- Fill ^{1st} row with 0

② Same algo as prev but ags kisi box mein negative value arhi to use ZERO kidega

③ for getting the actual string we will start from the highest value & then move with the arrows until we get 0.

		A	T	G	C	T
		0	0	0	0	0
A	0	1	0	0	0	0
G	0	0	0	1	0	0
C	0	0	0	0	2	0
T	0	0	1	0	0	3

GCT

seq 1

		A	C	G	C	T	G
		0	0	0	0	0	0
C	0	1	0	0	0	0	0
A	0	0	1	0	0	0	0
T	0	0	0	0	0	1	0
G	0	0	0	2	1	0	0
T	0	0	0	0	0	3	2

From this matrix we can get 3 global alignments.

P1: A C G C T G
- C - A T G T

P2: A C G C T G -
- C A - T G T

neche
upar se neche P1 work
path ka dekhte
ne aayenge agr
right par jayenge
to seq 1 ka element
ayega & seq 2 par
- Downwards
jayenge to seq 2
ka element ayega
& seq 1 mein -
diagonal main dono
k element honge.

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PS: $\begin{array}{c} \text{A}^{\text{C}}\text{GCTG} \\ \text{CATG_T_} \end{array}$

▷ SCORING MATRICES

- Previous scoring matrix mein hum ek fixed penalty value assign krskhe the pr each insertion deletion or indel or gap. (all are same thing)
- However, agar ek alignment mein 100 consecutive gaps hai to -100 tak penalty chahi jayegi, which is a severe penalty.

~~GATCCAG~~

~~GA_C_A_G~~
(low score)

- A series of k indels often represents a single evolutionary event (gap) rather than k events.

$\begin{array}{c} \text{G A T C C A G} \\ \text{G A _ C _ A G} \end{array}$
 $\begin{array}{cc} -1 & -1 \\ \textcircled{-2} & \end{array}$

$\begin{array}{c} \text{G A T C C A G} \\ \text{G A _ _ C A G} \end{array}$

iske instead of -2, hum 1st indel ko -1, and uske bad wale ko -0.5 dena chaahite hai.

that
 → Biologist considers these two events should have diff penalties

• PAM Matrix

↳ One of the types of scoring matrices

↳ Full Form: Point Accepted Mutation

↳ It is a matrix where each column and row represents one of the twenty standard amino acids

This matrix is regularly used as substitution matrices to score sequence alignments for proteins

↳ An accepted point mutation in protein is a replacement of one amino acid by another

• AFFINE GAP PENALTY

↳ One way to score gaps more appropriately is to define an affine penalty for a gap of length "k" as

$$\sigma + e \cdot (k-1)$$

where,

σ = gap opening penalty (penalty of 1st symbol of the gap)

e = gap extension penalty (penalty of the gap after 1st symbol)

k = length of gap

$$\sigma > e$$

$$\sigma = -5, e = -1$$

A G C T A C T

A - - - - C T

→ $k = 4$

$$\text{score} = -5 + (-1 \times 3) = -8$$

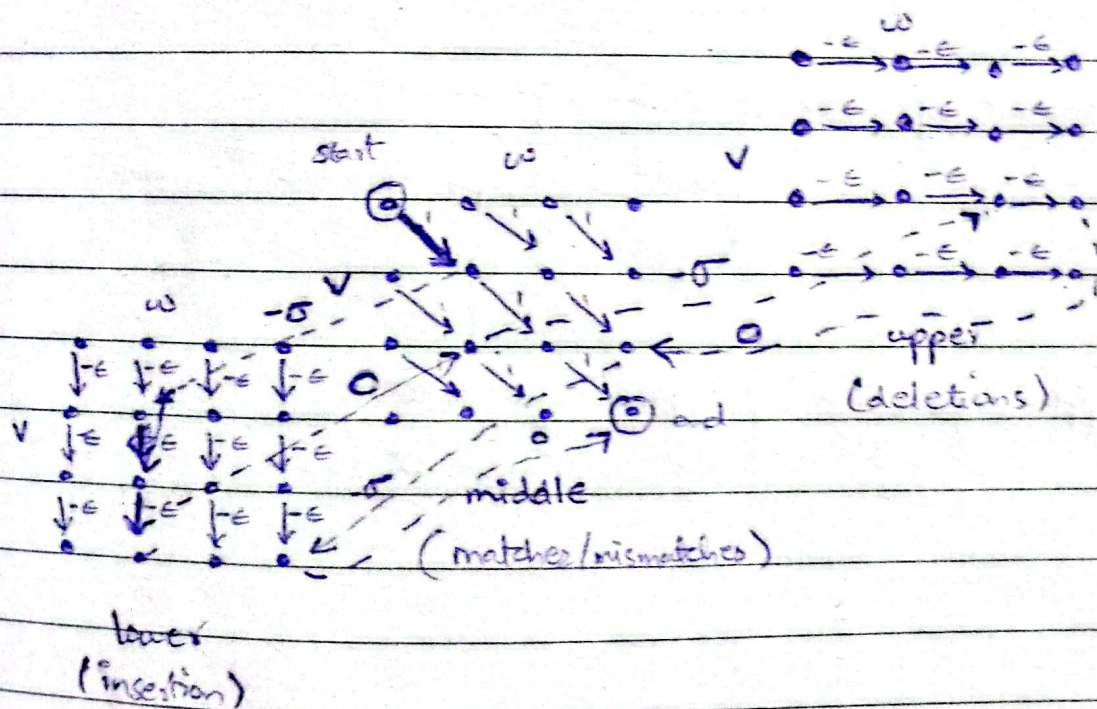
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↳ Ab agar har gap ki penalty -5 hoti to $-5 \times 4 = -20$ score hota but affine gap penalty ki waja se -9 hai.

↳ To model this problem into a DAG so that we can get score of 2 sequences thru recurrence relation.

↳ We build a three-level graph for alignment with affine gap penalties



→ Agar matches/mismatches dekhne honge to middle graph traverse krenge, agar downward jana hai (like we need to insert an element in v) to lower graph par shift kr lenge with -5 score or phir job take lower graph mein hai to $-e$ thega score.

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↳ Phir ags ko match/mismatch bata hai to usis middle mein jayenge with 0 cost.

↳ Isi tarique se ags delete krna hai to upper matrix mein jayenge.

$$\text{lower } i, j = \max \begin{cases} \text{lower } i-1, j - \epsilon \\ \text{middle } i-1, j - \delta \end{cases}$$

$$\text{middle } i, j = \max \begin{cases} \text{lower } i, j \\ \text{middle } i-1, j-1 + \text{score}(V_i, W_j) \\ \text{upper } i, j \end{cases}$$

$$\text{upper } i, j = \max \begin{cases} \text{upper } i, j-1 - \epsilon \\ \text{middle } i, j-1 - \sigma \end{cases}$$